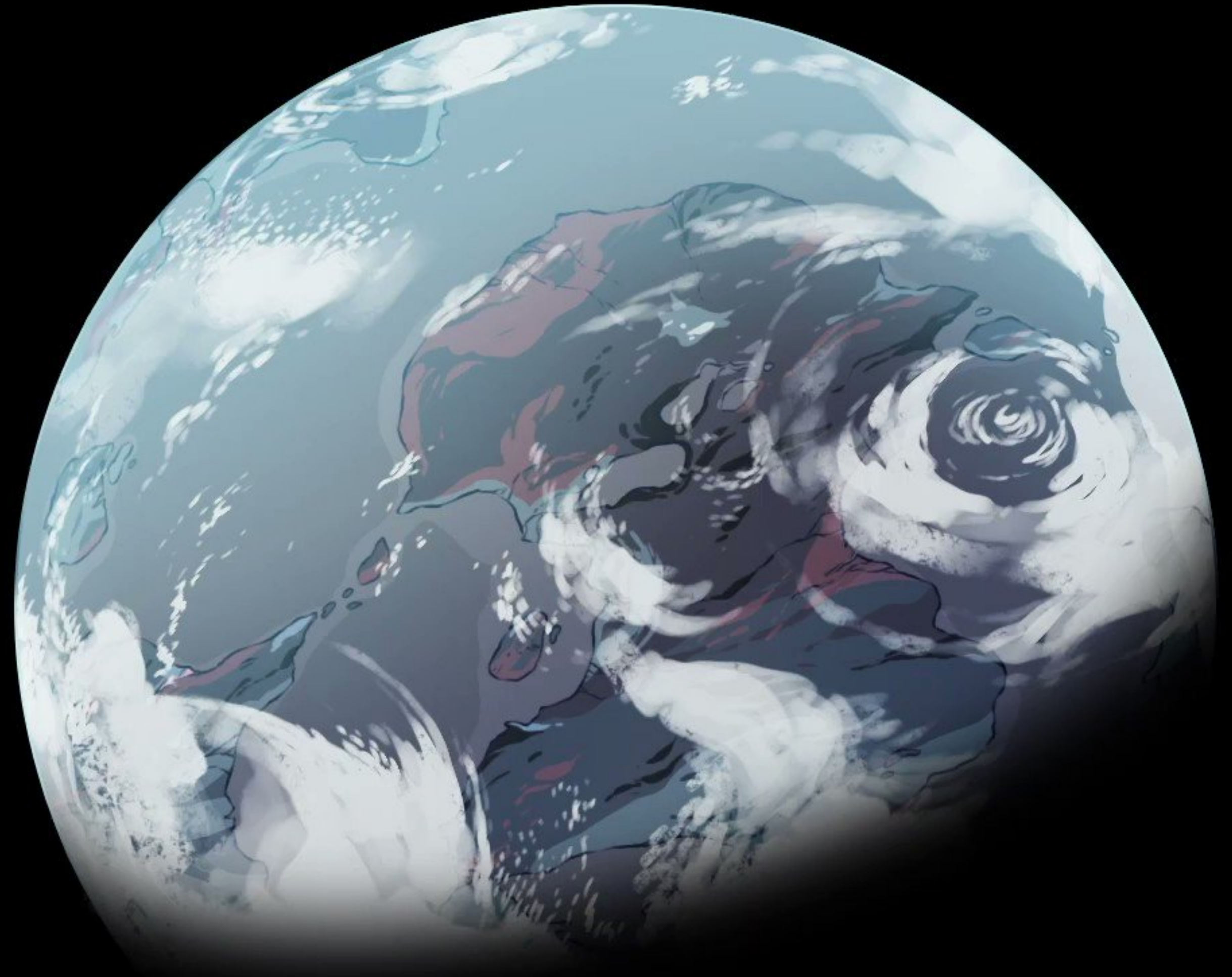


Science & Science Fiction

33-120 | Prof. Katie Breivik



Aims of the course

- a non-traditional approach to physical science credits
 - introduction of science concepts with primarily film & tv sci-fi source material
- engage with sci-fi source material through a variety of assignments, projects, and class discussions
- empower students to think critically about science concepts as they enter into everyday life

Aims of the course

Be able to discern whether science fiction topics are:

- Realistic
 - based on concrete science that is accurately reflected by the onscreen material
- Possible in the future or a *prediction of future technology*
 - could happen in a hypothetical situation but hasn't happened yet
- Impossible
 - physically impossible regardless of future technical developments

Course material

required textbook:

be sure to get the 2nd edition!

all film/tv clips will be shown in class

all assignments/assessments will be administered through Canvas

Office hours: 3pm on Tuesdays or by email!

Barry B. Luukkala

Exploring Science Through Science Fiction

Second Edition



 Springer

Course Component	Breakdown	Total Points
Science Knowledge Survey	1 on 1st day of class	25 pts
In-Class Discussion	Every class period	50 pts
Sci-fi Problem Sets	4 at 25 pts each	100 pts
Exploration Papers	5 at 25 pts each	125 pts
Quizzes on Lectures/Reading	7 w/ variable pts	350 pts
Team Projects	2 at 25 pts each	50 pts
Film Critiques	2 at 50 pts each	100 pts
Final Exam		200 pts
Total Points		1000 pts

Final grade boundaries are: A: 900 - 1000, B: 800 - 899, C: 700 - 799 pts D: 600 - 699, R: <600

Boundary cases may be discussed based on demonstrated engagement, effort, and improvement over the semester.

- Science knowledge survey: available now!
 - due @ 11:59pm tonight (01/13)
 - 25 pts; 30 mins to complete from starting time
 - 100% participation — I will update all attempted scores to 100% after deadline
- Quiz 1 on intro material will be available on Friday, 01/17
 - Due @ 11:59pm the same day
 - 20 pts; 30 mins to complete from starting time

Course policy notes

- **NO MAKEUP QUIZZES**
 - There will be several announcements of *extra credit opportunities* to make up for missed quizzes.
- Assignments can only be submitted once; be sure that you submit the version that you want to be graded
- Attendance will not be taken, but you must be in class to achieve discussion points (50 pts total)

Discussion questions

- 1.5 discussion pts per class period + 37 class periods
==> you can miss four classes and still get full points!
- *not quiz-style questions!*
 - questions are designed to be completely subjective so that they seed discussion — there are no right answers!
- answers collected using Google forms
 - you must sign in with your CMU andrew gmail address to fill in the form

**You do not need to email me unless you will miss more than
4 classes over the semester**

Discussion question 1: <https://tinyurl.com/SciSciFi-1>



Week	Monday	Wednesday	Friday
Week 1	01/13/25 Course overview and early works of sci-fi Critique of the first sci-fi film Imagining human spaceflight in 1902 Science knowledge survey (25 pts)	01/15/25 Reference frames, acceleration and artificial gravity	01/17/25 What is the nature of space and time? Classical physics and Newton's laws Quiz 1: Intro material (20 pts)
Week 2	01/20/25 No Class: Martin Luther King Jr Day	01/22/25 Newton's laws continued Einstein and modern physics	01/24/25 Einstein and modern physics Relativistic spacetime Problem 1: acceleration (25 pts)
Week 3	01/27/25 Einstein and modern physics Special relativity and time travel	01/29/25 General relativity Black holes and gravitational waves	01/31/25 No Class: Spacetime Team Project Problem 2: time dilation (25 pts)
Week 4	02/03/25 General relativity continued Gravitational waves Exploration Paper 1: Space & Time (25 pts)	02/05/25 The warp drive and "Future physics": Quantum gravity Spacetime Team Project (25 pts)	02/05/25 "Future physics": Quantum gravity, worm holes, and the Multiverse Hypothesis
Week 5	02/10/25 What is the Universe made of? The standard model of particle physics Problem 3: General relativity (25 pts)	02/12/25 Dark matter and dark energy Quiz 2: Space and Time (85 pts)	02/14/25 Nuclei, atoms, and gases
Week 6	02/17/25 Liquids, solids, phase transitions	02/19/25 Optical properties; transparent materials	02/21/25 Metamaterials and invisibility cloaks Problem 4: matter & materials (25 pts)
Week 7	02/24/25 Chemical energy Midterm Film Critique (50 pts)	02/26/25 Nuclear weapons; matter & antimatter	02/28/25 Can a machine become self aware? Information storage & data processing Exploration Paper 2: Matter & Energy (25 pts)

🚨🚨🚨 Extra Credit Opportunity! 🚨🚨🚨



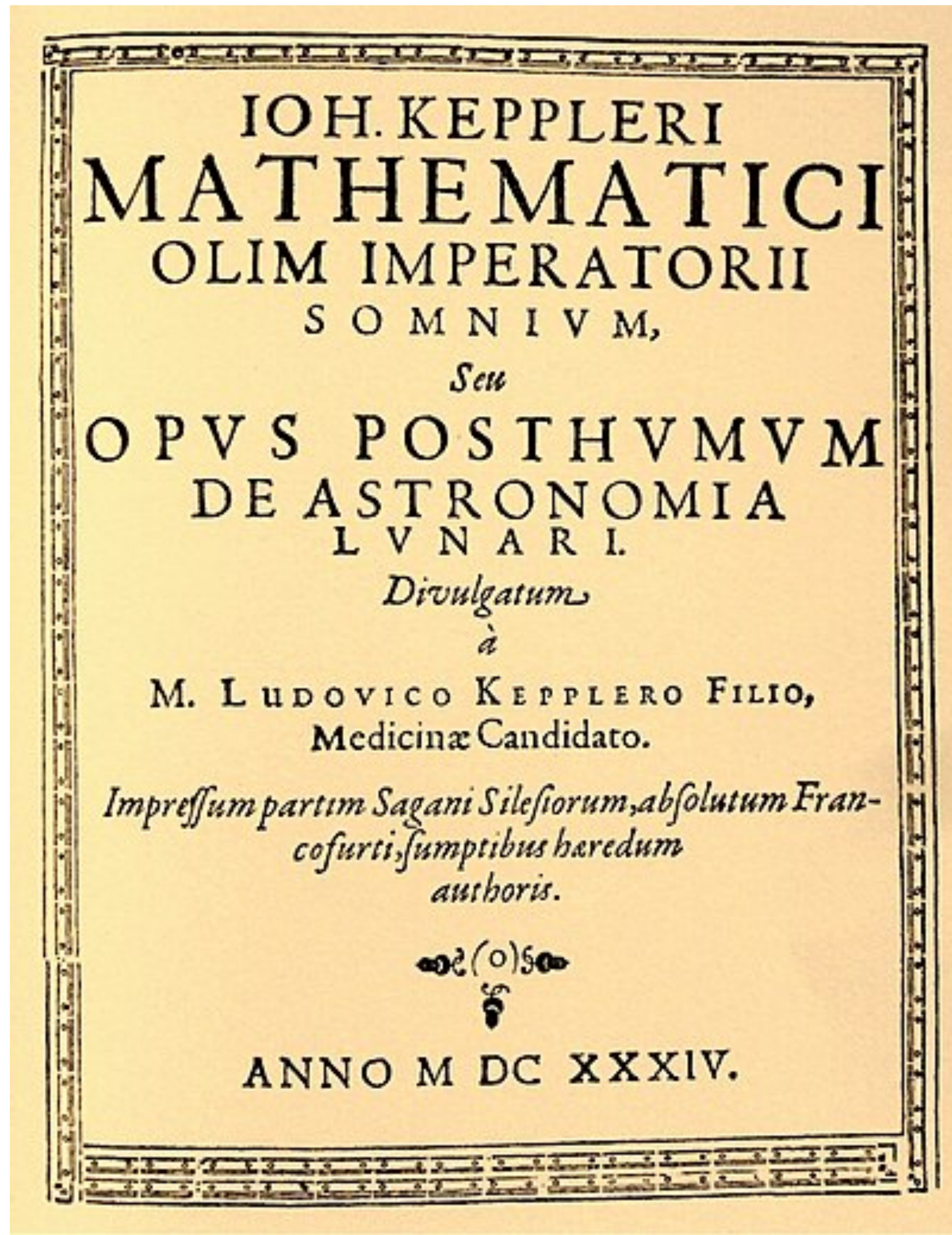
<https://tinyurl.com/SciSciFi-EC-1>

Due before class on Wednesday

Tell me what you are most
interested in covering this
semester!

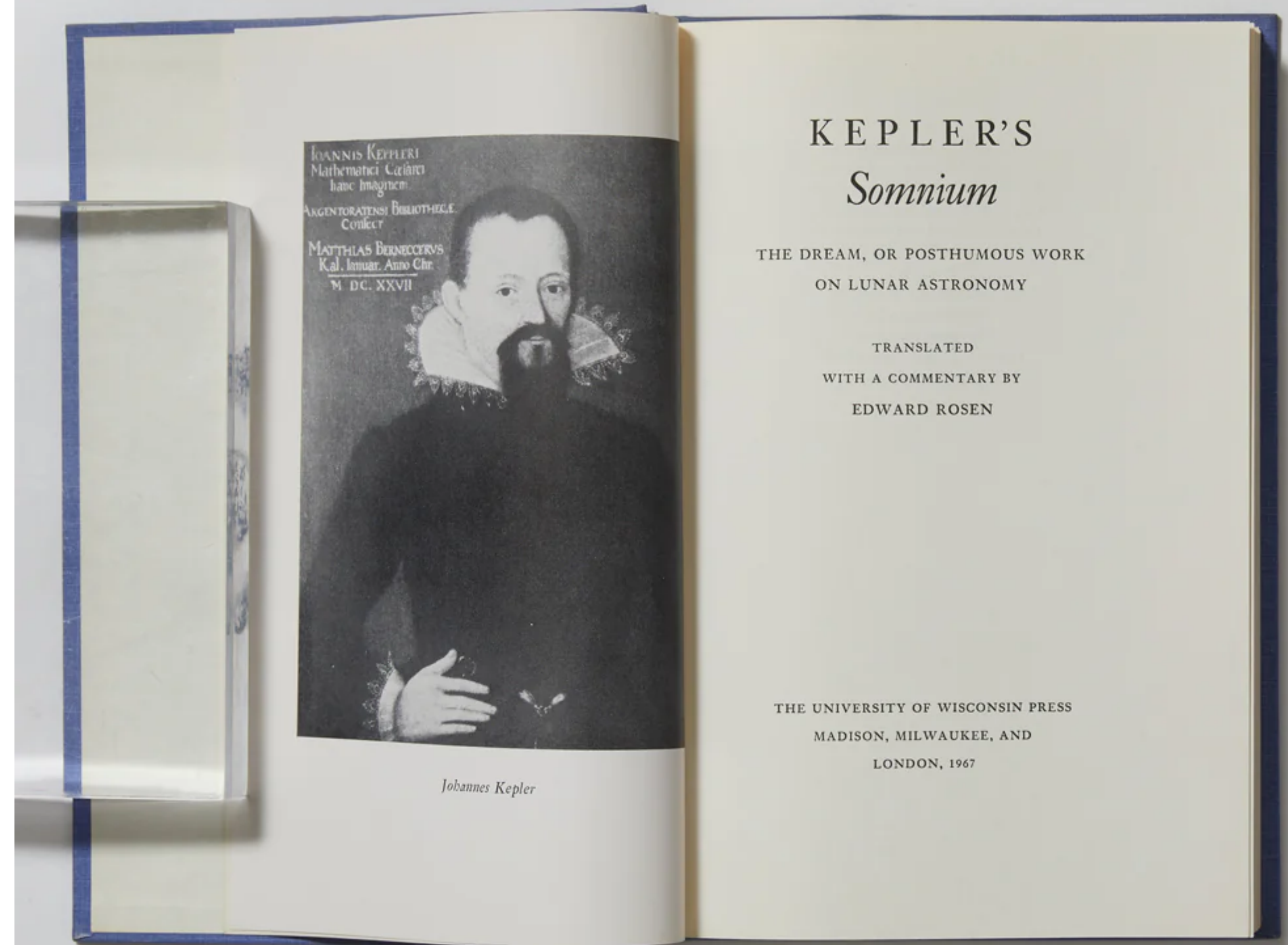
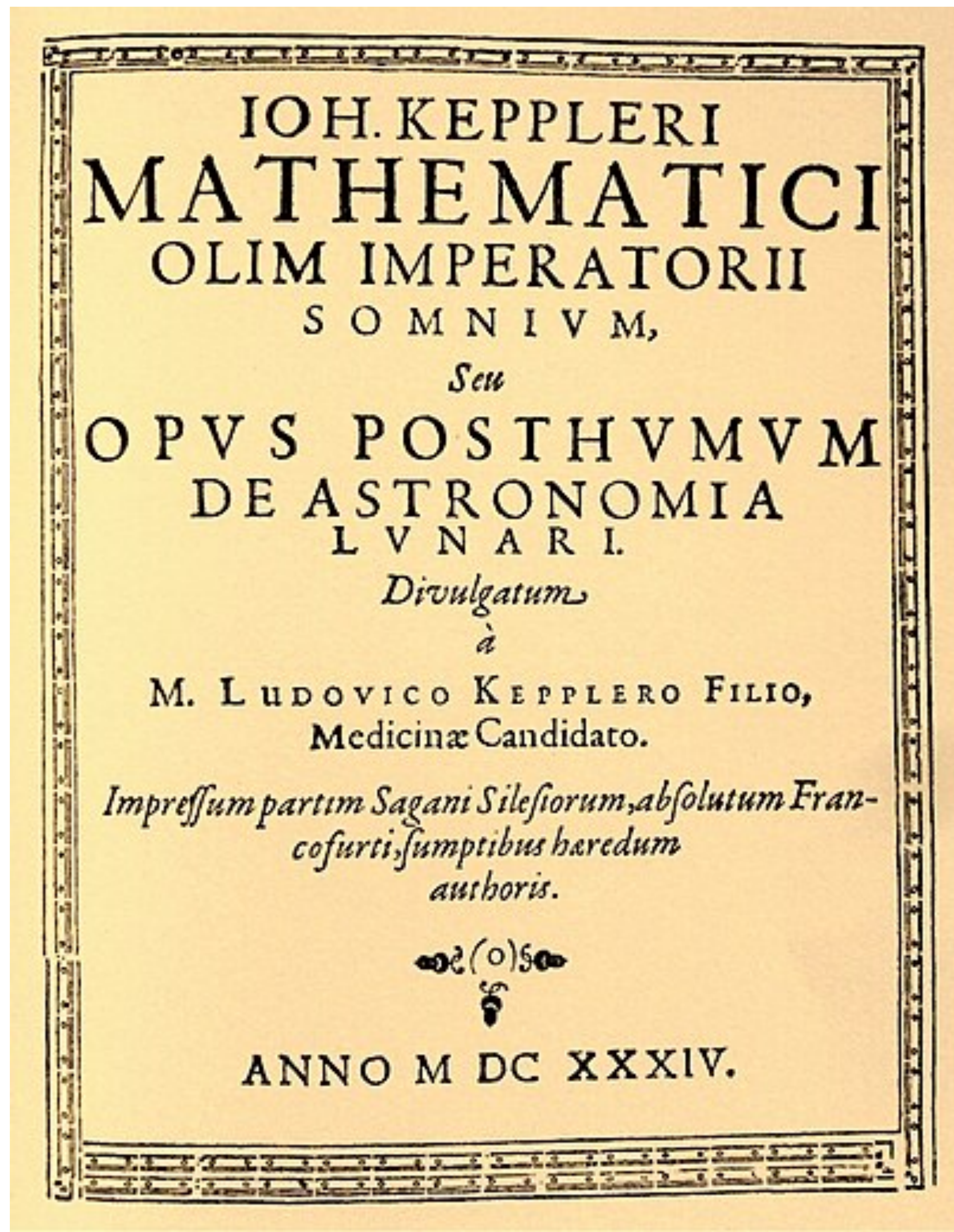
The
earliest
works
of
science
fiction

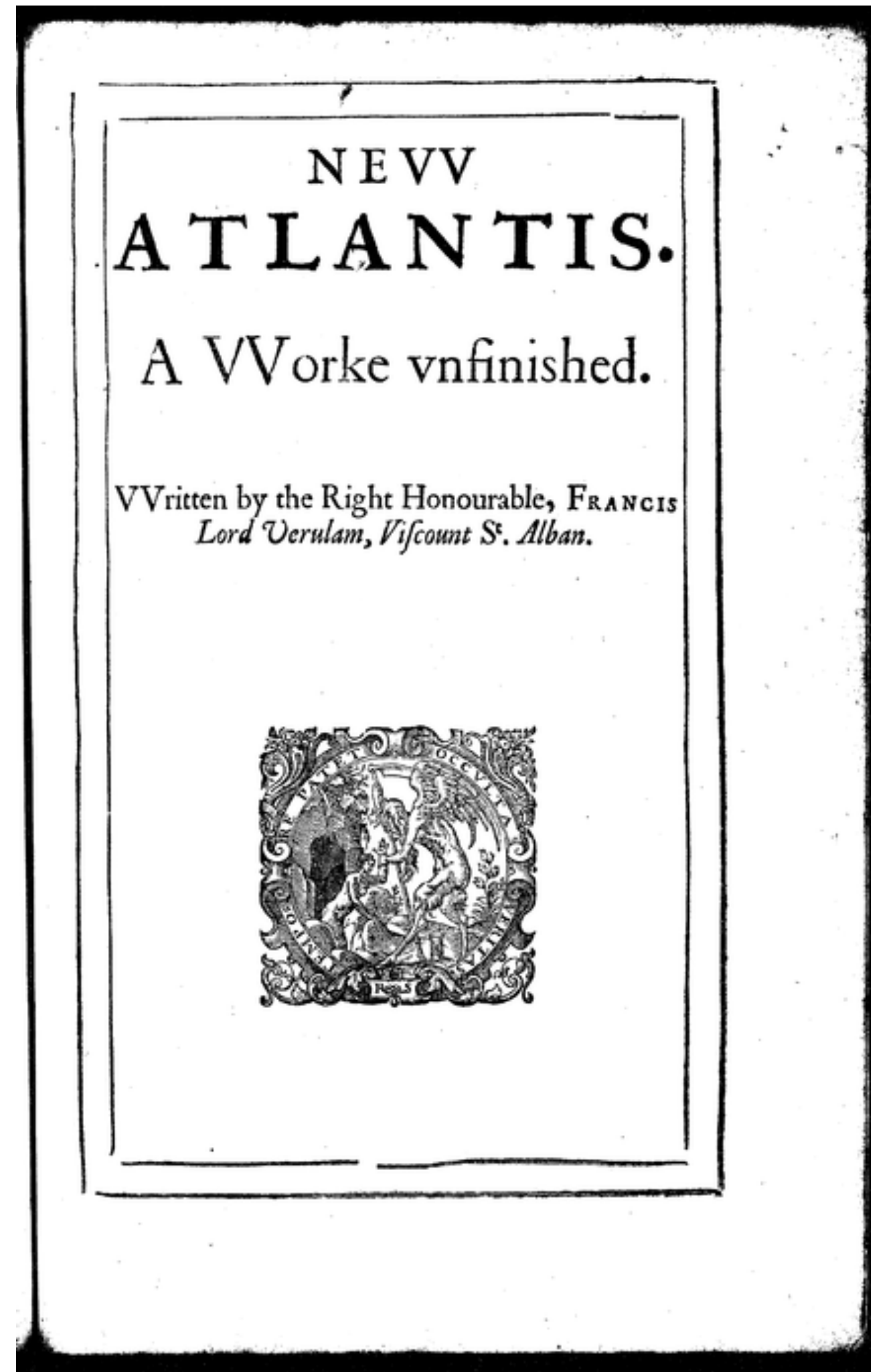




First published and finished
literary work of science fiction

Published posthumously in 1634





Some argue that Francis Bacon's Atlantis should be counted

Published in 1624, but unfinished

A Trip to the Moon

Georges Méliès (1902)



Le Voyage Dans La Lune: a critique

Realistic

- Bullets and guns are real
- Seeing the sky really bright makes sense without light pollution
- Rocket shape
- Moon is very cold
- Craters
- Casting barrel of gun
- Moon people are light because of less gravity

Possible/Prediction

- Rocket landing in the water
- With enough power the angle of the barrel was possible
- Retrieving floating rocket

Impossible

- Bullets will not make it to the moon
- No astronaut
- Umbrellas $\neq$ mushrooms
- Rocket would not just fall back to earth
- Women didn't go to the moon first/moon people in general
- Forests on the moon

Next time:

- ▶ inertial reference frames
- ▶ Newton's laws

Reminder:

- Science knowledge survey: available on Canvas
 - due @ 11:59pm tonight (Jan 13)
 - 25 pts; 30 mins to complete from starting time

